	GLOBAL ECONOMICS GRADE: 10TH TERM 3 — LEARNING EVIDENCE 4 WEB IMPLEMENTATION AND REPORT OF CONTINUOUS PROBABILITY MODELING (C8–C10) TEACHER'S NAME: Nicolás López Cuéllar	THIRD TERM 2025–2026

Learning objective: Create interactive computational tools for modeling continuous probability distributions by graphing PDFs, calculating probabilities under curves, estimating parameters from data, comparing fitted distributions with empirical histograms, and writing an evidence-based report using real-life data.	**Assessed criteria:** C8: Creates an interactive webpage that simulates continuous probability distributions and calculates probabilities under the PDF. C9: Uses AI-assisted programming to create a tool that models data by estimating distribution parameters and comparing fitted PDFs with empirical histograms. C10: Uses continuous probability models and real-life data to write an evidence-based report with statistical results, contextual interpretation, and APA-style references.

Background from C1–C7

In C1–C4, you modeled continuous random variables with linear and piecewise probability density functions (PDFs), ensuring non-negativity and total area equal to 1. In C5–C7, you worked with the normal distribution, standard-normal calculations, and inverse-normal reasoning. In this activity, you will extend that work into a computational project that combines probability modeling, visualization, data analysis, and evidence-based reporting.

Task overview

Create one webpage (HTML + CSS + JavaScript) that performs continuous probability modeling in two connected modes, and use it to support a short written report:

1. **Simulation mode (C8):** users modify distribution parameters, graph PDFs, and calculate probabilities under the curve.
2. **Data modeling mode (C9):** users upload or enter data, estimate distribution parameters, and compare fitted PDFs with empirical histograms.
3. **Report work (C10):** students use real-life data and real-world sources to write a short evidence-based report.

The page must be functional, not only visual. Do not submit solved exercises, completed answer sheets, or screenshots as a replacement for functionality.

GitHub and deployment tutorials

Use these tutorials if you need help preparing and publishing your webpage:

1. Tutorial to download Git and push a project into GitHub.
2. Tutorial to host a website using GitHub Pages.

Use the following CSV datasets to complete the activity:

- Dataset 1.
- Dataset 2.
- Dataset 3.

Required AI conversation links

For the work to be valid, the deployed webpage must include a visible section titled **AI development record** or **AI conversations used**. This section must contain hyperlinks to the AI conversation or conversations used to help create the webpage. These links must be accessible directly from the deployed webpage, not only from the written report.
If the deployed webpage does not include these AI conversation links, the C8–C9 implementation will be considered incomplete.

Mandatory requirements for C8

Your C8 implementation must include interactive PDF graphs for the following distributions:

1. Uniform distribution.
2. Triangular distribution.
3. Increasing or decreasing linear distribution.
4. Piecewise distribution.
5. Normal distribution.

For each distribution, your webpage must:

1. Allow the user to enter or modify all defining parameters.
2. Generate the corresponding PDF from the parameters.
3. Display the PDF graph interactively.
4. Compute probabilities under the curve for:
   - probability below a point,
   - probability above a point,
   - probability between two points.
5. Visually shade the selected probability region.
6. Display the numerical probability value.
7. Update the graph and values whenever parameters or bounds change.
8. Include a hyperlink to the webpage deployed through GitHub Pages or another accessible public/shared link.
9. Include a visible **AI development record** section in the deployed webpage with hyperlinks to the AI conversation(s) used.

Distribution-specific C8 conditions

Your webpage must satisfy the following conditions:

1. **Uniform:** input lower endpoint and upper endpoint; generate a constant PDF.
2. **Triangular:** input lower endpoint, upper endpoint, and mode; generate a triangular PDF.
3. **Linear:** define an increasing or decreasing linear PDF on a finite interval; ensure total area under the PDF is 1.

4. **Piecewise:** define at least one piecewise PDF using intervals and formulas; graph all pieces correctly.

5. **Normal:** input mean and standard deviation; graph the normal PDF.

Mandatory requirements for C9

Your C9 implementation must use AI-assisted programming to create a tool that receives data and estimates distribution parameters.
The tool must support at least the following fitted models:

1. Uniform distribution.

2. Triangular distribution.

3. Linear distribution.

4. Piecewise uniform distribution.

5. Normal distribution.

Your webpage must allow the user to upload or input data and choose which distribution to fit. For each fit, display:

1. estimated parameters,

2. fitted PDF,

3. histogram of uploaded data,

4. histogram and fitted PDF on the same graph,

5. short interpretation of whether the fitted distribution appears reasonable,

6. a hyperlink to the webpage deployed through GitHub Pages or another accessible public/shared link.

7. visible hyperlinks in the deployed webpage to the AI conversation(s) used.

Required documentation of AI use for C9

You may use AI tools, but you must document your process. Submit:

1. a deployed webpage link using GitHub Pages or another accessible public/shared link,

2. prompt(s) used,

3. AI-generated code or code fragments,

4. visible AI conversation hyperlinks inside the deployed webpage,

5. short explanation of how AI was used,

6. short explanation of what you modified, verified, or corrected,

7. short explanation of how you tested correctness.

Required test datasets for C9

Use these provided CSV files to validate your C9 tool:

- Download Grade 10 dataset 1.

- Download Grade 10 dataset 2.
- Download Grade 10 dataset 3.

These datasets are simulated for educational purposes and are provided to test the C9 data-modeling mode. Your webpage must be able to upload each file, display the histogram, estimate or use the required distribution parameters, graph the fitted PDF, and calculate probabilities from the fitted model. You may use additional datasets, but these three files must work correctly in your webpage.
Include screenshots or equivalent evidence in the C10 report showing that each provided CSV file was tested in your C9 webpage.

Mandatory requirements for C10

Your C10 work must be a short report based on continuous probability modeling and real-life data. The report must follow APA style and include all of the following:

1. A title.
2. A real-life context connected to the data.
3. At least **three real-world sources**.
4. A clear research question.
5. Description of the data used, including how each provided CSV file was uploaded in the webpage.
6. Hyperlink to the deployed webpage (GitHub Pages or another accessible public/shared link).
7. Explanation of the fitted probability models.
8. Estimated parameters for each fitted model.
9. Probability calculations from the fitted models.
10. Comparison of the fitted models using graph evidence and probability results.
11. Screenshots showing the provided CSV files being used in the webpage.
12. A short section referencing the AI conversation links included in the deployed webpage.
13. A contextual conclusion based on the modeling results.
14. APA-style in-text citations when using information from the sources.
15. A references section in APA format.

C10 modeling requirements

For the C10 report, analyze real-life data and complete the following:

1. Use the three provided CSV datasets to model continuous data, and you may add real-life datasets as additional evidence.
2. For each provided dataset, display the dataset histogram.
3. For each dataset, fit at least three continuous probability models.
4. At minimum, include:
   - piecewise uniform model,
   - linear model,

- normal model.

5. Plot each fitted PDF on the same graph as the histogram.

6. Compare visually which model appears to fit better.

7. Explain one strength and one limitation of each model for the dataset.

C10 probability-calculation requirements

For each dataset and each fitted model, calculate:

$$P(\mu - \sigma < X < \mu + \sigma)$$

and

$$P(X < \mu - 2\sigma \text{ or } X > \mu + 2\sigma).$$

Use probabilities from the fitted model, meaning area under the model PDF, not only point-counting from the raw data.

APA format requirements for C10

The report must use APA style in the following way:

1. Use clear academic headings, such as title, introduction, methodology, results, discussion, conclusion, and references.

2. Use author–year in-text citations when making contextual claims. Example: `(Author, year)`.

3. Include a references section titled **References**.

4. List the references in alphabetical order by the first author's last name or by the institutional author.

5. Each reference must include the author or institution, year, title, source name, and URL or DOI when available.

6. Use APA 7th edition style for web sources. Example format: Author or Institution. (Year). *Title of page or report*. Website or Publisher. URL

7. Do not include sources in the references section if they are not cited inside the report.

Real-world source requirements for C10

The three real-world sources must satisfy the following:

1. They must be connected to the context of the report.

2. They must be reliable sources, such as official statistics sites, academic publications, institutional reports, or reputable data portals.

3. They must be cited inside the report when making contextual claims.

4. They must appear in the references section.

Deliverables and evaluation checklist by criterion

Submit and verify the following items:

1. **C8: Simulation webpage mode**

   - Submit `index.html` or an equivalent main HTML file.

- Submit linked stylesheet file(s) and JavaScript file(s).
- Submit the hyperlink to the webpage deployed through GitHub Pages or another accessible public/shared link.
- The deployed webpage must include visible hyperlinks to the AI conversation(s) used.
- The webpage must include all required distribution types.
- The webpage must allow parameter changes for each distribution.
- The PDF graph must update when parameters are modified.
- The shaded probability region must update when bounds are modified.
- The numerical probability must update correctly.
- The webpage must work for at least one example of each required distribution type.

2. **C9: Data modeling webpage mode**

   - Submit the AI-assisted parameter estimation script or the webpage code section that performs the estimation.
   - Submit the hyperlink to the webpage deployed through GitHub Pages or another accessible public/shared link.
   - The deployed webpage must include visible hyperlinks to the AI conversation(s) used.
   - Validate the webpage with the three provided datasets (dataset 1, dataset 2, dataset 3). Additional datasets are allowed, but the provided files must work correctly.
   - Submit AI-use documentation, including prompts, AI-generated code or fragments, how AI was used, what you verified or corrected, and tests.
   - Uploaded or entered data must be processed successfully.
   - Estimated parameters must be displayed.
   - The histogram must appear on the same graph as the fitted PDF.
   - The fit interpretation must be consistent with the displayed graph.

3. **C10: Real-life data report**

   - Submit the C10 report as a PDF or source document.
   - Include the hyperlink to the deployed webpage.
   - Submit or clearly reference the real-life data used in the report.
   - Submit or clearly reference at least three real-world sources.
   - The report must include a research question, data description, methodology, model fitting results, probability calculations, model comparison, discussion, conclusion, and references.
   - The report must include screenshots of the provided CSV files being used, including histograms and fitted PDFs.
   - The report must briefly explain how the CSV data was uploaded in the webpage.
   - The report must include a short section referencing the AI conversation links included in the deployed webpage.
   - The report must calculate $P(\mu - \sigma < X < \mu + \sigma)$ for each fitted model.
   - The report must calculate $P(X < \mu - 2\sigma \text{ or } X > \mu + 2\sigma)$ for each fitted model.
   - The report must compare models using graph evidence and probability results.
   - The report must use APA-style in-text citations and APA-style references.
   - The conclusion must be consistent with the data and fitted models.